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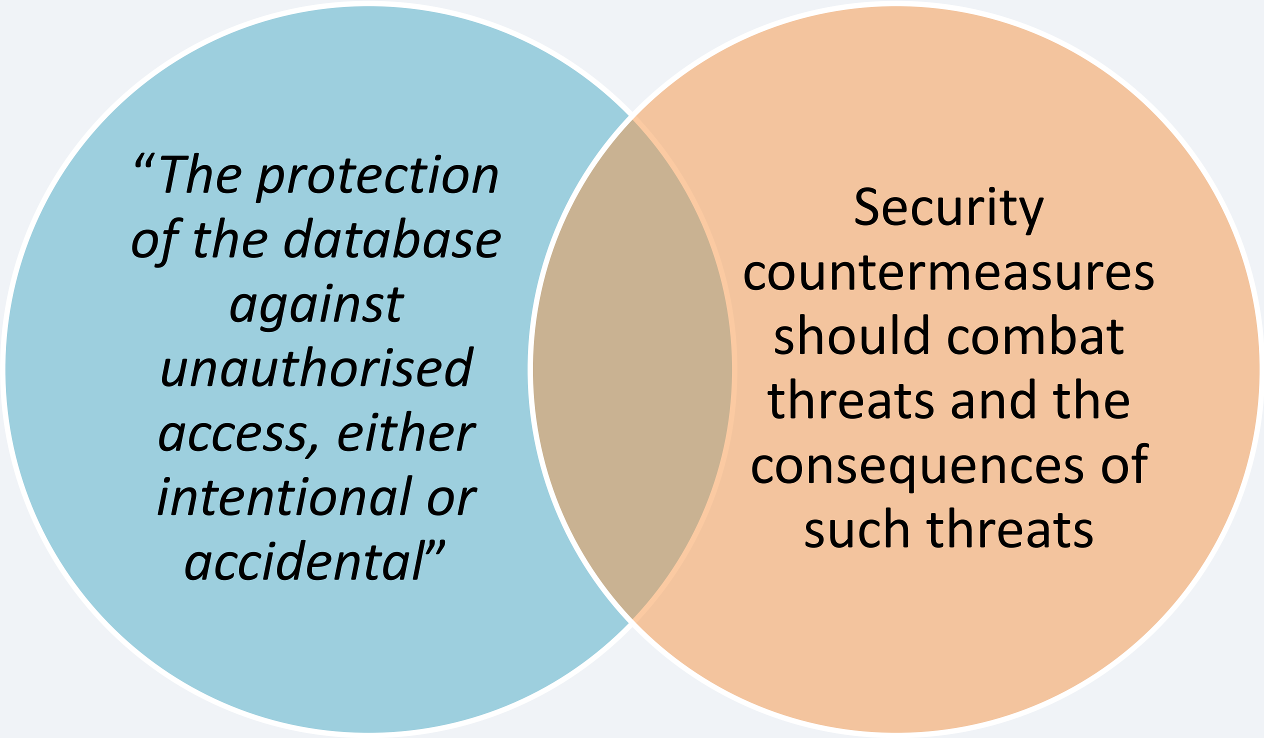
COM 736

Database Systems and Data Analytics (Week 7 Additional Materials)

# Database Security – An Overview

# Database Security



# Security Threats

**Hardware**
- Fire/flood/bom bs
- Power loss/surge
- Theft of equipment

**Software**
- Program alteration
- Theft of programs
- SQL Injection
- Malicious software

**Administrators**
- Inadequate security
- Excessive Privilege

**Networks**
- Wire tapping
- Cable breaking

**Database**
- Unauthorised access
- Theft of Data
- Data corruption

**Users**
- Using another user's access
- Unauthorized access
- Hacking
- Blackmail
- Viruses

**Operators**
- Trapdoors
- Alterations
- Inadequate training
- Inadequate security

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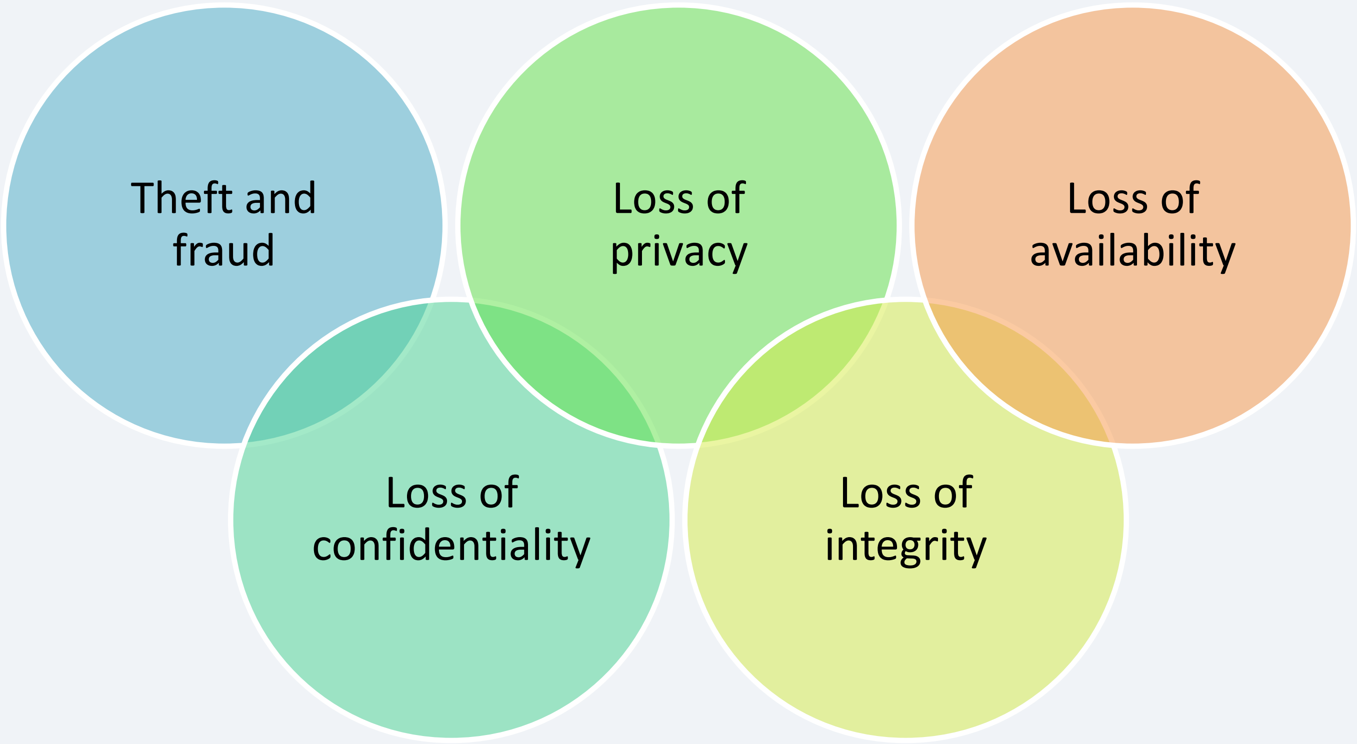
# Compromising Security



# Countermeasure

| | | |
|---|---|---|
| **Authorization** | **Integrity** | Views |
| Backup and Recovery | Encryption | RAID |

# Authorization

Access controls

Granting a right or privilege to access a specific system or system object

User or role based access

Select, update, insert, delete data

# Integrity

Constraints imposed at an object level to control the data being inserted

| Entity integrity | Referential integrity | Domain integrity | Semantic integrity |
|---|---|---|---|
| No attribute of a primary key can be null | A candidate row exists within a related table | Allowable values for a given attribute | Ensuring data adheres to business rules |

# Views

A means of restricting user access to certain data

Can restrict to specific attributes within a table

More restrictive way of maintaining access control at a sub-table level

# Encryption

Encoding of data by algorithm,  data will be unreadable without decoding

Symmetric key encryption

- e.g. DES (Data Encryption Standard)

Asymmetric key encryption

- e.g. RSA

Sometimes can be used in together

# Backup and Recovery

Addresses threats such as hardware errors

Logging tracks the changes to the database

Making sure that there are reliable Backup routines in place is very important for the data recovery

Most of the data is generated through transactions

Support smooth data recovery after a hardware/storage failure

# RAID

- Creates a more fault tolerant system than using a single disk
- Redundant Array of Independent Disks
- Copies data across a bank of disks
- Different RAID levels can be used addressing redundancy (mirroring), striping and parity
- Addresses similar issues to backup and recovery

# Next

Database security measures

DBMS provides various security measures to protect the database systems and the data